

Lab: Policy Selection and Active Inference Loop

Learning Goal: Implement and analyze multi-step policy selection with EFE decomposition.

Part 1: Generate and Inspect Policies

```
policies = generate_policies(num_actions=3, policy_length=2)
print(f"Total policies: {len(policies)}")
for i, p in enumerate(policies):
    print(f"   $\pi_{\{i\}}$ : {p}")
```

How does the number of policies scale with `num_actions` and `policy_length`?

Write your response here

Part 2: EFE Decomposition

Exercise: Using a simple 2-state, 2-action model, compute EFE for each policy and decompose:

```

A = np.array([[0.9, 0.1], [0.1, 0.9]])
B = np.zeros((2, 2, 2))
B[:, :, 0] = np.eye(2) # stay
B[:, :, 1] = np.array([[0, 1], [1, 0]]) # switch
C = np.array([2.0, -1.0])
qs = np.array([0.5, 0.5]) # uncertain

policies = generate_policies(2, 1)
for i, p in enumerate(policies):
    G, prag, epist = compute_efe_decomposed(A, B, C, qs, p)
    print(f"π={p}: G={G:.3f}, Pragmatic={prag:.3f}, Epistemic={epist:.3f}")

```

Which policy wins and why?

Write your response here

Part 3: Run the Full Loop

Run the active inference loop for 20 steps and analyze:

```

log = active_inference_loop(A, B, C, D, env, num_steps=20, gamma=1.0)
print(f"Action frequencies: {np.bincount(log['actions'], minlength=2)}")

```

Does the agent converge on the preferred state?

Write your response here

Part 4: Gamma Sweep

Run with gamma = 0.1, 1.0, and 10.0. Compare action selection behavior.

Write your response here

Part 5: Reflection

In 150 words, reflect: How does watching the EFE decomposition evolve over time help you understand the agent's decision-making? When was epistemic value highest vs. pragmatic?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Policy generation	Combinatorial policy space
2	EFE analysis	Pragmatic vs. epistemic decomposition
3	Loop execution	Full active inference cycle
4	Parameter sweep	Gamma and decisiveness
5	Behavioral analysis	Understanding agent decisions